Compare function types

1 The table of values for function P and function Q are provided:

- Function P

x	-2	-1	0	1	2
y	9	6	3	0	-3

- Function Q

x	0	1	2	3	4
y	6	3	2	3	6

a Determine what type of functions Function P and Function Q are.

b Sketch the graph of the functions on the same coordinate plane.

c As $x \to \infty$, determine which function will have the greater value.

d Determine if each function is increasing or decreasing on the domain $x < 2$.

2 Consider the following pair of functions:

- Function P
  $y = -12 + \dfrac{x}{10}$

- Function Q

x	-8	-7	-6	-5	-4
y	-1	-4	-5	-4	-1

a Sketch the graph of the functions on the same coordinate plane.

b How many times do P and Q intersect?

c Determine which function has the greater function value at the y-intercept.

d Determine which function has the greater function value at $x = -1$.

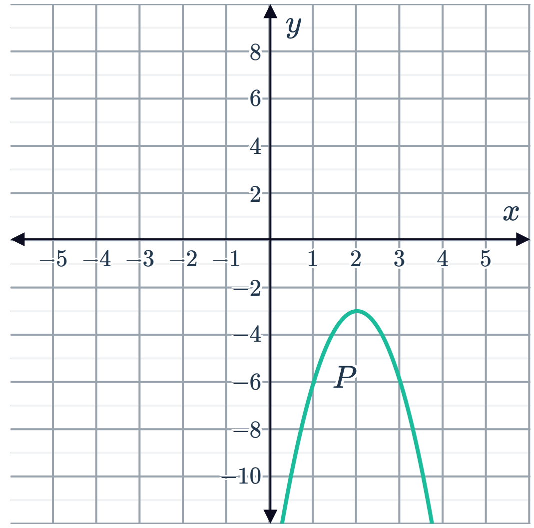
3 The line P is given by $y = -4 + \dfrac{4x}{3}$ and the parabola Q is given by $y = -(x-1)(x-4)$.

a Sketch the graphs of line P and the parabola Q on the same coordinate plane.

b Identify how many times P and Q intersect.

c Identify which function has the greater function value at $x = 0$.

4 The graph of function P, given by the equation $y = -3\left(x-2\right)^2 - 3$ is shown:



a Sketch the graph of function Q given by the equation $y = -6x + 12$ on the same coordinate plane.

b How many times do P and Q intersect?

c Determine which function has the greater function value at the y-intercept.

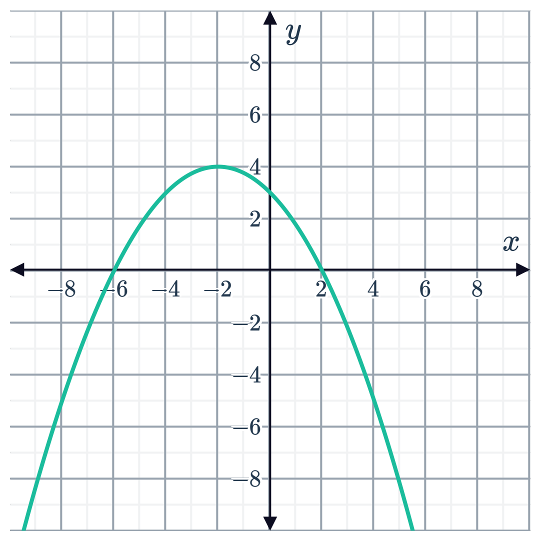
d Determine if each function is increasing or decreasing on the domain $x < 2$.

5 Determine which quadratic function has the greatest maximum value:

- Function P

x	-1	1	3
y	2	6	2

- Function Q



6 A table of values for parabola P and for parabola Q are provided:

- Parabola P

x	-8	-4	0
y	4	2	4

- Parabola Q

x	0	1	2
y	-5	-4	-5

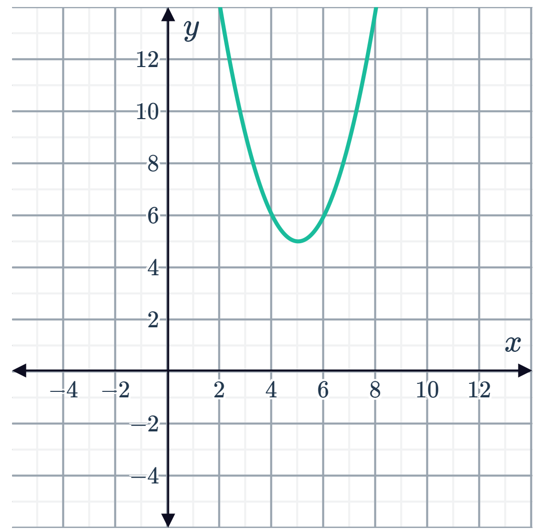
a Sketch the graph of parabolas P and Q on the same coordinate plane.

b Identify which parabola is concave up.

c Identify which parabola has the greater function value at the y-intercept.

7 Consider the following pair of functions:

- Function A
  $y = -2(x-7)(x-3)$
- Function B



a Determine which function has a maximum of $y = 8$.

b Determine which function has a minimum of $y = 5$.

c Determine how many zeros function A has.

d Determine how many zeros function B has.

8 Parabola P is given by $y = \dfrac{11(x-6)^2}{36} - 6$ and parabola Q is given by $y = \dfrac{13(x-3)^2}{9} - 8$.

a Sketch the graph of parabolas P and Q on the same coordinate plane.

b Determine which function has the lesser minimum.

c Identify how many times the parabolas intersect.

d Determine which parabola has the greater function value at $x = 8$.

9 Cubic function P is given by $y = -(x+2)^3 + 4$ and cubic function Q is given by $y = (x-1)(x-2)(x+2)$.

a Sketch the graph of cubic functions P and Q on the same coordinate plane.

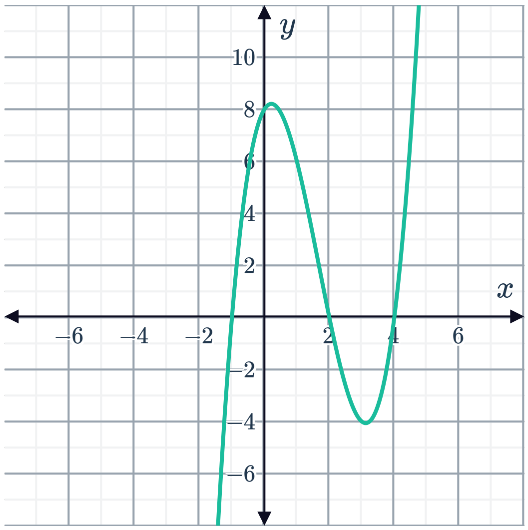
b Determine which cubic function has the greater function value at $x = -1$.

c Determine how many zeros cubic function P has.

d Determine how many zeros cubic function Q has.

10 Consider the following pair of cubic functio

- Function P



- Function Q

x	-5	-4	-3	-2
y	0	-6	-4	0

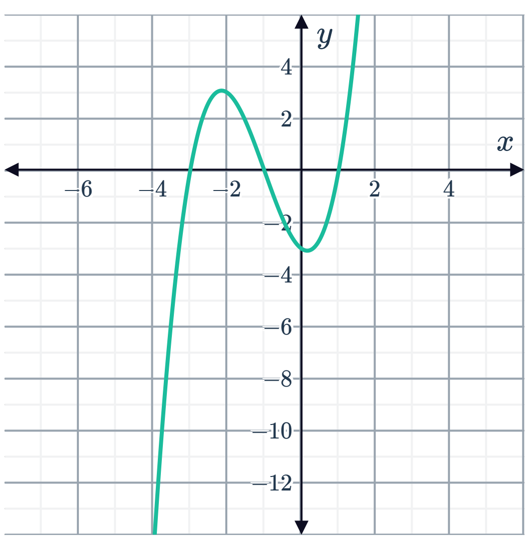
a Sketch the graph of function Q on the same coordinate plane as function P.

b Determine if each function is increasing or decreasing on the domain $1 < x < 3$.

c Describe the end behavior of each exponential function as $x \to \infty$.

11 Consider the following pair of cubic functions:

- Function P



- Function Q
  $y = -(x+6)(x+2)(x+1)$

a Sketch the graph of function Q on the same coordinate plane as function P.

b Determine which function has the greater function value at the y-intercept.

c Determine if each function is increasing or decreasing on the domain $-2 < x < -2$.

d Describe the end behavior of each exponential function as $x \to \infty$.

12 A table of values for the cubic function P and for the cubic function Q are provided:

- Cubic function P

x	-5	-4	-3	-2
y	0	6	4	0

- Cubic function Q

x	-1	0	1	2
y	0	9	8	3

a Sketch the graph of cubic functions P and Q on the same coordinate plane.

b Determine which cubic function has the greater function value at the y-intercept.

c Determine how many zeros cubic function P has.

d Determine how many zeros cubic function Q has.

13 The exponential function J is given by $y = 9\left(3^{-x}\right)$.
The following table shows the function values for exponential function K:

x	-1	1	2	5	10
y	15	$\frac{5}{3}$	$\frac{5}{9}$	$\frac{5}{243}$	$\frac{5}{59\,049}$

a Determine if the exponential function K is increasing or decreasing.

b Sketch the graph of exponential functions J and K on the same coordinate plane.

c Describe the end behavior of each function as $x \to \infty$.

d Determine what transformation we can apply to the function $y = 3^{-x}$ to produce function K.

14 The exponential function E is given by $y = 3^x + 4$ and the exponential function F is given by $y = -8^x + 1$.

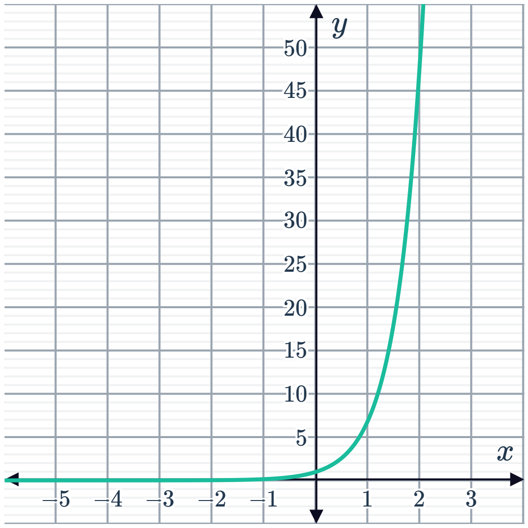
a Sketch the graph of exponential functions E and F on the same coordinate plane.

b Identify how many times E and F intersect.

c Describe the end behavior of each exponential function as $x \to \infty$.

15 Consider the following pair of functions:

- Function P



- Function Q
  $y = 2^x + 4$

a Determine how many times the graphs of P and Q intersect.

b Determine which function tends to infinity the quickest as $x \to \infty$.

16 Consider the following pair of functions:

- Function P
  $y = 6^{-x} + 5$

- Function Q

x	-4	-3	-2	-1	0
y	0	$-\frac{11}{4}$	$-\frac{11}{2}$	$-\frac{33}{4}$	-11

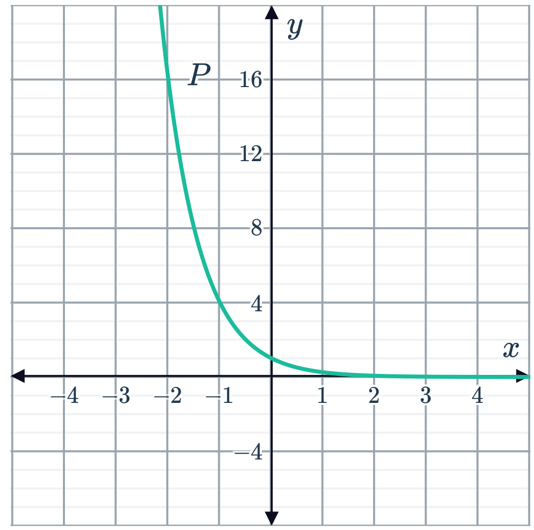
a Determine what type of functions Function P and Function Q are.

b Sketch the graph of the functions on the same coordinate plane.

c Identify which function has the greater function value at the y-intercept.

d Identify how many times P and Q intersect.

17 The graph of the exponential function P, given by $y = 4^{-x}$ is shown on the right:



a Sketch the graph of function Q given by the equation $y = 8x + 12$ on the same coordinate plane.

b How many times do P and Q intersect?

c Which function has the greater function value at the y-intercept?

d Which function has the greater function value at $x = 1$?

18 The exponential function P is given by the equation $y = 5^{-x} + 3$ and the linear function Q is given by the equation $y = -4x + 4$.

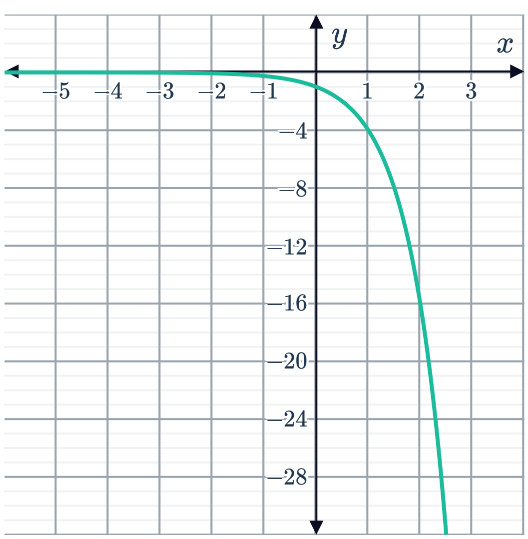
a Sketch the graph of the functions on the same coordinate plane.

b How many times do P and Q intersect?

c As x approaches ∞, which function is greater?

19 Consider the following pair of functions:

- Function A



- Function B

$$y = -\frac{4(x-3)(x-7)}{21}$$

a Determine which function has the greater function value at the y-intercept.

b Determine if each function is increasing or decreasing on the domain $0 < x < 5$.

20 The exponential function P is given by $y =$ [illegible] 4 and the parabola Q is given by $y = \dfrac{(x-6)(x-2)}{4}$.

a Sketch the graph of exponential function P and the parabola Q on the same coordinate plane.

b Determine which function has the greater function value at $x = 4$.

c Determine if each function is increasing or decreasing on the domain $1 < x < 3$.

21 The table of values for the functions P and Q are provided:

- Function P

x	-7	-6	-5	-4	-3	-2
y	-7	-4	-3	-4	-7	-12

- Function Q

x	-10	-5	-1	1	5	10
y	$1\,073\,741\,824$	$32\,768$	8	$\frac{1}{8}$	$\frac{1}{32\,768}$	$\frac{1}{1\,073\,741\,824}$

a Determine what type of functions Function P and Function Q are.

b Sketch the graph of the functions on the same coordinate plane.

c Determine how many times functions P and Q intersect.

Applications

22 A table of values for the function P and a table of approximate values for the function Q is provided below:

- Function P

x	-4	-3	-2	-1	0	1	2
y	-3	0	3	6	9	12	15

- Function Q

x	-5	-2	-1	0	1	5	10
y	$-32\,765$	-61	-5	2	$2\frac{7}{8}$	$2\frac{32\,767}{32\,768}$	$2\frac{1\,073\,741\,823}{1\,073\,741\,824}$

a Which of the functions are linear and non-linear?

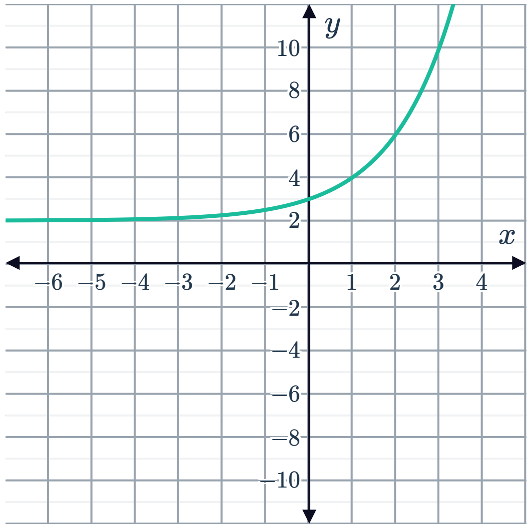
b Sketch the graph of function P.

c Given that Q is an exponential function, sketch the graph of the function Q on the same coordinate plane.

d Using part (c), how many solutions are there to the equation $P - Q = 0$?

23 Consider the graph of the exponential funct[illegible] and the table of values for parabola Q:

- Function $P : y = 2^x + 2$



- Parabola $Q : y = -3\left(x+5\right)\left(x+2\right)+2$

x	-5	-4	-3	-2	-1
y	2	8	8	2	-10

a. Sketch the graphs of P and Q on the same coordinate plane.

b. Describe the behavior of function P and function Q as x approaches infinity.

c. Using the graph, determine how many solutions there are to the equation $2^x + 2 + 3\left(x+5\right)\left(x+2\right) - 2 = 0$.

24 The population of two different bacteria, labeled J and K, are given by the following table of values:

Bacteria J:

t (time in days)	0	1	2	3	4
P (population)	1	60	3600	2.16×10^5	1.296×10^7

Bacteria K:

t (time in days)	0	1	2	3	4
Q (population)	1	50	2500	1.25×10^5	6.25×10^6

a. Determine which population of bacteria increases faster.

b. The population P of bacteria J at time t can be modeled using the standard equation $P\left(t\right) = a^t$, where $a > 1$. By using the table of values, graph $P\left(t\right)$ for $t > 0$.

c. The population Q of bacteria K at time t can be modeled using the standard equation $Q\left(t\right) = b^t$, where $b > 1$. By using the table of values, graph $Q\left(t\right)$ for $t > 0$.